Definitions

1 Machine Learning

1.1 Hypothesis Space

The set of all functions a model can represent, denoted by $\mathcal{H}$.

$$\mathcal{H} \subseteq \{h : \mathcal{X} \to \mathcal{Y}\}$$

Here, h is a hypothesis mapping inputs $\mathcal{X}$ to outputs $\mathcal{Y}$.

1.2 Inductive Bias

The assumptions that guide a learning algorithm toward certain hypotheses, allowing it to generalize beyond its training data.

Restriction bias. Limits the hypothesis space $\mathcal{H}$ to a smaller set of possible functions.

$$\mathcal{H}_{\text{restricted}} \subseteq \mathcal{H}$$

A smaller hypothesis space can require less data to generalize, provided it contains a sufficiently good hypothesis.

Preference bias. Favors certain hypotheses over others without necessarily excluding them from $\mathcal{H}$.

For example, weight decay encourages solutions with smaller weights.

1.3 L2 Norm

The L2 norm, also known as the Euclidean norm, measures the magnitude (length) of a vector.

For a vector $w \in \mathbb{R}^n$:

$$\|w\|_2 = \sqrt{\sum_{i=1}^{n} w_i^2}$$

For example, given $w = [3, 4]$:

$$\|w\|_2 = \sqrt{3^2 + 4^2} = 5$$

The squared L2 norm removes the square root:

$$\|w\|_2^2 = \sum_{i=1}^{n} w_i^2$$

1.4 L2 Regularization

A regularization technique that discourages large model weights, helping reduce overfitting.

Instead of minimizing only prediction loss, we add a penalty proportional to the squared L2 norm of the weights.

$$L_{\text{total}} = L_{\text{prediction}} + \frac{\lambda}{2}\|w\|_2^2$$

Where:

- $L_{\text{prediction}}$ is the original loss function.
- w is the vector of model weights being regularized.
- $\lambda \geq 0$ controls regularization strength.
- $\|w\|_2^2$ is the sum of squared weights.

Larger values of λ penalize large weights more strongly. When $\lambda = 0$, there is no regularization.

Effect on Gradient Descent

The penalty contributes an additional term to the gradient:

$$\frac{\partial L_{\text{total}}}{\partial w} = \frac{\partial L_{\text{prediction}}}{\partial w} + \lambda w$$

Substituting into the gradient descent update:

$$w_{\text{new}} = w - \eta \left(\frac{\partial L_{\text{prediction}}}{\partial w} + \lambda w \right)$$

Rearranging:

$$\boxed{w_{\text{new}} = (1 - \eta\lambda)w - \eta \frac{\partial L_{\text{prediction}}}{\partial w}}$$

Thus, each update shrinks the weights by a factor of $(1-\eta\lambda)$ before applying the prediction-loss gradient.

This is equivalent to weight decay in standard SGD.